

Slow shell building, a possible trait for resistance to the effects of acute ocean acidification

George G. Waldbusser,^{*1} Matthew W. Gray,² Burke Hales,¹ Chris J. Langdon,² Brian A. Haley,¹ Iria Gimenez,¹ Stephanie R. Smith,¹ Elizabeth L. Brunner,¹ Greg Hutchinson²

¹College of Earth, Ocean, and Atmospheric Sciences, Oregon State University, Corvallis, Oregon

²Coastal Oregon Marine Experimental Station and Department of Fisheries and Wildlife, Hatfield Marine Science Center, Oregon State University, Newport, Oregon

Abstract

Increasing anthropogenic carbon dioxide is altering marine carbonate chemistry through a process called ocean acidification. Many calcium carbonate forming organisms are sensitive to changes in marine carbonate chemistry, especially mollusk bivalve larvae at the initial shell building stage. Rapid calcification, limited energy reserves, and more exposed calcification surfaces, are traits at this stage that increase vulnerability to ocean acidification through our previously argued kinetic-energetic hypothesis. These developmental traits are common to broadcast spawning bivalve species that are the focus of most ocean acidification studies to date. Some oyster species brood their young, which results in slower development of the embryos through the initial shell formation stage. We examined the responses of the brooding Olympia oyster, *Ostrea lurida*, during their initial shell building stage. We extracted fertilized eggs from, *O. lurida*, prior to shell development, then exposed developing embryos to a wide range of marine carbonate chemistry conditions. Surprisingly, *O. lurida* showed no acute negative response to any ocean acidification treatments. Compared to the broadcast spawning Pacific oyster, *Crassostrea gigas*, calcification rate and standardized endogenous energy lipid consumption rate were nearly 10 and 50 times slower, respectively. Our results suggest that slow shell building may lessen the energetic burden of acidification at this stage and provides additional support for our kinetic-energetic hypothesis. Furthermore, these results may represent an example of exaptation; fitness conveyed by a coopted trait that evolved for another purpose, a concept largely lacking in the current perspective of adaptation and global climate change.

The combustion of fossil fuels has led to a rapid rise in atmospheric carbon dioxide (IPCC 2014) that exceeds in magnitude and rate any observed in the last million years (Lüthi et al. 2008), leading to the subsequent acidification of the world's oceans (Doney et al. 2009). Some ocean ecosystems are more sensitive to these changes in ocean carbonate chemistry due to conditions already near laboratory defined organismal thresholds (under otherwise ideal conditions), such as upwelling zones (Harris et al. 2013), or due to other factors altering carbonate chemistry such as eutrophication (Waldbusser et al. 2011). Ultimately, it appears that increasing atmospheric carbon dioxide is additive to systems that appear already enhanced with carbon dioxide (Feely et al. 2010; Hauri et al. 2013; Takahashi et al. 2014). While this bodes unfavorably for the many calcium carbonate forming organisms found globally, it is now clear that many organisms may have discrete periods of heightened vulnerability

to ocean acidification due to various life-history characteristics (Waldbusser et al. 2010; Byrne 2011; Waldbusser and Salisbury 2014). For example, embryological development is a period of heightened sensitivity to a variety of stressors, and is one of the motivations for the use of bivalve embryos in toxicological studies (His et al. 1996). The period of embryological development and first shell formation for commercially grown oysters has been well documented as a bottleneck for hatchery production of oysters in response to ocean acidification (Barton et al. 2012; Waldbusser et al. 2013).

The U.S. Pacific Northwest shellfish industry has been particularly hard hit from ocean acidification (Barton et al. 2015), and the larvae of the non-native Pacific oyster, *Crassostrea gigas*, grown in this region appear highly vulnerable to this stressor. The greatest sensitivity occurs during the organogenesis of the first shell, called the prodissoconch I (PDI), due to the rapid calcification rate, limited endogenous energy, and an apparent greater exposure of calcification

*Correspondence: waldbuss@coas.oregonstate.edu

surfaces to the external environment (Waldbusser et al. 2013, 2015a). These characteristics are shared among most broadcast spawning bivalves and broadcast spawning species have been the primary focus of most ocean acidification studies to date (Gazeau et al. 2013). In contrast, brooding species of oysters develop far more slowly during PDI formation (days instead of hours) (Hopkins 1936; Baker 1995). If the rapid rate of calcification is central to larval bivalve sensitivity to ocean acidification (as in Waldbusser et al. 2013), slow shell development in brooding oysters may provide one possible mechanism for the recent findings of increased resistance of brooded invertebrate larvae to global change (Noisette et al. 2014; Lucey et al. 2015). Indeed, the limited studies to date however suggest brooded larvae maybe more resilient to acidification than broadcast spawned larvae, with important caveats seen in negative carry over effects from the swimming veliger stage to adult stages (Hettinger et al. 2012). In general however, studies have generally focused on post-release larvae in brooding oyster species and have not removed possible maternal effects during brooding.

The Olympia oyster (*Ostrea lurida*) is native to a large portion of the North American Pacific coast, and broods internally fertilized eggs for 10–12 d before release as veliger larvae, typically having a shorter veliger duration than broadcast spawned larvae (Hopkins 1936; Baker 1995). The brooding *O. lurida* produces fewer and larger eggs than the broadcast spawning *Crassostrea gigas* (Jaekle 1995), as is expected in cases of increased maternal investment. These differences in life-history strategies and rates of development between a broadcast and a brooding oyster species provides an opportunity to explore whether slow shell development during embryogenesis may provide some resilience in an acidifying ocean, and thus a possible example of exaptation. An alternative hypothesis is that since the geographic origin of both species is different, simply being adapted to habitats of differing carbonate chemistry conditions may prove to explain differences in responses of larvae.

The primary focus on organismal resiliency to ocean acidification from a physiological perspective has been the ability to alter acid-base chemistry of internal or semi-internal bodily fluids (Portner 2008; Melzner et al. 2009). Analogous to thermal tolerances, organisms are to some degree, adapted to physiological ranges of carbonate chemistry reflective of their environment (Portner 2008), and some organisms show acclimation potential by altering internal acid-base chemistry over a period of days while others from more stable environments appear to not have the same acclimation potential (Pane and Barry 2007). This perspective has provided a key framework for understanding global change impacts on various species through range expansion/contraction (in the case of temperature) (Phillips et al. 2010; Wetthey et al. 2011; Foden et al. 2013) or through greater plasticity in acid-base regulation related to habitat variability in carbon dioxide

(Pane and Barry 2007). Fitness is however conveyed through traits and while these perspectives encompass adaptation potential of specific traits to a changing climate, the potential for exaptation has largely been ignored (but see Sammarco and Strychar 2009). A trait that originally conveyed fitness for a different purpose may be coopted for fitness in a new way through exaptation (Gould and Vrba 1982). In other words, some species may already possess traits that convey resiliency that could be used in novel ways in light of the rapidly changing conditions due to human effects on global climate. A focus on strict deterministic links between genetic variance and optimal physiological adaptation to local conditions cannot therefore anticipate the full potential of life-history strategies and traits that organisms already possess which may provide fitness to ocean acidification.

To examine the response of slow shell building larvae to acidification during embryological development, we extracted 1-d old embryos from female *O. lurida* to verify the timing and feasibility of culturing larvae outside the brood chamber. We then determined shell development and growth responses of pre-shelled, 48 h old *O. lurida* embryos to the partial pressure of CO₂ (P_{CO_2}) and aragonite saturation state (Ω_{ar}) independently across 16 treatments (Fig. 1; Table 1) through carbonate system manipulation (following Waldbusser et al. 2015a,b). As in our previous studies pH varies pseudo-independently, in other words, we do not have complete orthogonality of pH with P_{CO_2} and Ω_{ar} , but there are multiple similar pH values at different P_{CO_2} and Ω_{ar} levels that provide inferential power to examine responses to pH (see lines of equal pH in Fig. 1). Finally, we documented the timing and rate of calcification and changes in standardized lipid composition over the course of development and growth of the PDI shell in both *C. gigas* and *O. lurida*.

Methods

Carbonate chemistry exposure experiments

Larval collection

In May 2013 (Experiment 1) and April 2014 (Experiment 2), 72 adult *O. lurida* oysters from Oregon Oyster Farms (Yaquina Bay, Oregon) were transferred to a flow-through conditioning trough at the Hatfield Marine Science Center (HMSC), Newport, OR. Oysters were continuously fed a mixture of microalgae (*Chaetoceros gracilis* and *Isochrysis galbana*) ad libitum. Seawater temperatures were raised from 15°C to 20°C at a rate of 1°C d⁻¹. Experiments were initiated by checking daily for a minimum of six brooding females for the appearance of extruded eggs, which occurred on 2th June 2013 (Experiment 1) and 10th May 2014 (Experiment 2). Due to the time needed to set up these experiments, extraction of larvae did not begin until 30–48 h after the brooding females with extruded eggs were identified. Three of the six females were sacrificed and the developmental stage of broods determined. The third sacrificed female from both

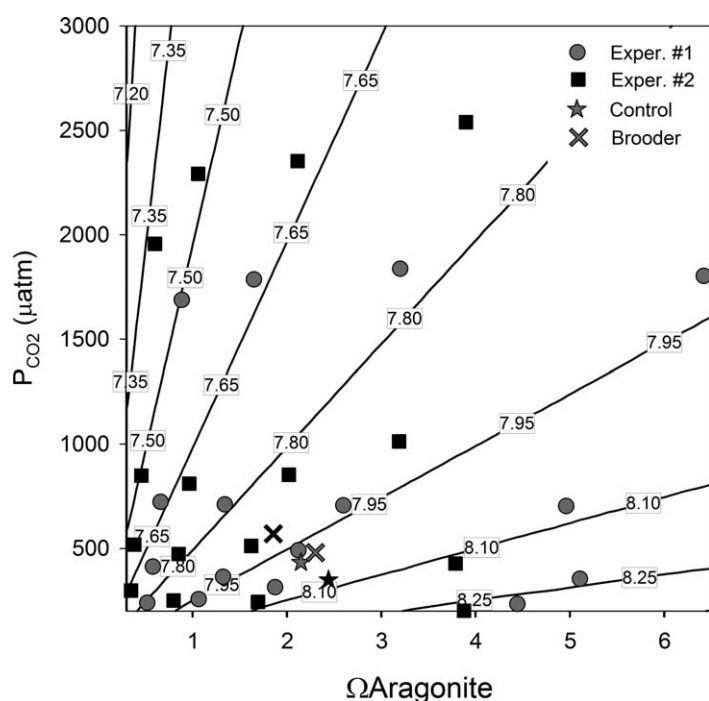


Fig. 1. Carbonate chemistry treatment values in saturation state and P_{CO_2} space for *O. lurida* response experiments, gray for experiment #1 and black for #2. Lines are isopleths of pH (total). Control values were conditions in the open culture containers, and the brooder values are the average conditions brooding oysters were maintained in parallel with the experiments.

experiments was either found to be brooding larvae that were further along in development (Experiment 1) or not to be brooding larvae at all (Experiment 2). The remaining three females were reared under filtered, flow-through control conditions to compare larval development under brooded and extracted conditions. No food was provided to brooding females to provide an equal comparison to non-brooded larvae in the experimental chambers. The two broods at the gastrulation stage were rinsed from the female's gills with 0.22 μm -filtered seawater and combined into a single 200 mL beaker partially filled with similar seawater. The suspension of embryos was then passed through a 140 μm sieve to remove large debris (e.g., gill tissue), re-suspended in filtered seawater, and then the remaining seawater decanted carefully with a 35 μm filter on a gentle siphon to prevent loss of larvae. The embryos were re-suspended again, and the suspension was sub-sampled to determine embryo density. Photographs of embryos were taken to document size and developmental stage (described below).

Chemistry manipulations

Carbonate chemistry manipulations were conducted following procedures detailed in Waldbusser et al. (2015a,b). Briefly, 1 μm -filtered seawater collected from Yaquina Bay, OR, was acidified with trace metal reagent grade 34–37%

HCl to almost completely titrate the alkalinity, then bubbled with outside air for 72 h to outgas most of the dissolved inorganic carbon (DIC) as CO_2 . The de-carbonated water was then 0.22 μm -filtered, added to sterilized 20 L polycarbonate carboys, pasteurized at 90°C for 3 h, and stored at 2–5°C until utilized. Prior to manipulations, the seawater was aerated with 0.22 μm -filtered ambient air for at least 24 h at room temperature. DIC and alkalinity values were calculated for all 16 treatment combinations of P_{CO_2} and Ω_{ar} targeted in the 4x4 factorial design at the experimental temperature and salinity conditions. A stock solution of Na_2CO_3 and NaHCO_3 (alkalinity of $\sim 3.3 \times 10^5 \mu\text{eq kg}^{-1}$ and DIC $\sim 2 \times 10^5 \mu\text{mol kg}^{-1}$) and 0.1 N certified HCl was used to adjust the final alkalinity and DIC concentrations for the various treatments. Treatment waters were stored in gas impermeable bags at 4°C without air head-space for less than 1 week prior to use.

Discrete carbonate chemistry samples for P_{CO_2} and DIC analysis were collected from the treatment water bags and from the control water just before stocking embryos in 500 mL biological oxygen demand (BOD) bottles. Samples were also taken from each biological oxygen demand (BOD) bottle and open bucket (the experimental containers described below) at the end of the incubation period by siphoning water through a 37 μm Nitex screen to prevent removal of larvae. During the entire incubation, a daily discrete carbonate chemistry sample was taken from the inlet to the flow-through brooding female incubation aquaria. All carbonate chemistry samples were collected in 350 mL amber glass bottles, preserved with 30 μL of saturated HgCl_2 , and sealed with polyurethane-lined metal crimp caps until processed. Analyses were carried out following the procedure described in Bandstra et al. (2006), modified for discrete samples by Hales et al. (2005). P_{CO_2} and DIC standards that bracketed the experimental range, which were larger than typical seawater values, were employed to ensure accuracy. Complete carbonate chemistry values were computed from P_{CO_2} and DIC using Millero (2010) carbonic acid dissociation constants with temperature and salinity dependencies (which capture the Lueker et al. (2000) seawater constants), Dickson (1990) constants for boric acid, and Millero (1995) water dissociation constants.

Experimental setup and sampling of *O. lurida* larvae

Three replicate 500 mL (BOD) bottles per treatment were used for the larval incubations and were filled with manipulated water without head-space. Two separate controls were also run: (1) BOD bottles filled with unaltered 1 μm -filtered water collected immediately before the de-carbonation process that were stored at 2–5°C until use, and (2) open culture buckets filled with 1 μm -filtered seawater that was freshly collected the day before the beginning of the experiment, bubbled with CO_2 -stripped air for 20 h, then bubbled with ambient air for 4 h ("Control" Figs. 1, 3). Embryos were

Table 1. Carbonate chemistry conditions for experiments and controls by experiment. Left two columns are the treatment categories for saturation state (Ω) and PCO_2 . All concentrations are in $\mu\text{mol kg}^{-1}$, and P_{CO_2} is in μatm . Temperature for both experiments was 20°C and average salinity was 28.9 and 28.3 for experiments #1 and #2, respectively. Avg. Brooder conditions were the average of daily discrete samples from the flow-through seawater system, and Development values are the average conditions in the open experimental containers in which both species were tracked during development. Control conditions were freshly collected and filtered seawater in open culture containers. Bolded treatment names lacked triplicate larval sub-samples.

Ωtrt	CO_2trt	Talk	DIC	HCO_3	CO_3	pH(total)	P_{CO_2}	Ωar
<i>Experiment #1</i>								
Low	Low	749	681	640	32	7.81	239	0.53
Low	MidLow	995	936	886	36	7.72	413	0.58
Low	MidHi	1367	1321	1255	41	7.63	723	0.67
Low	High	2352	2332	2220	55	7.51	1688	0.89
MidLow	Low	1136	1021	946	66	7.95	257	1.07
MidLow	MidLow	1476	1352	1258	82	7.93	364	1.33
MidLow	MidHi	1975	1875	1768	83	7.78	710	1.35
MidLow	High	3347	3276	3115	101	7.63	1786	1.66
MidHi	Low	1688	1514	1388	116	8.03	314	1.88
MidHi	MidLow	2169	1996	1848	131	7.96	491	2.13
MidHi	MidHi	2828	2637	2453	160	7.93	706	2.60
MidHi	High	4828	4656	4398	196	7.77	1836	3.21
High	Low	2499	2127	1845	274	8.28	234	4.45
High	MidLow	4064	3711	3382	305	8.07	703	4.97
High	MidHi	3158	2763	2437	313	8.23	354	5.11
High	High	7010	6628	6176	391	7.93	1802	6.43
Control		2073	1894	1747	132	7.99	433	2.15
Avg. Brooder		2194	1997	1836	145	7.98	480	2.31
<i>Experiment #2</i>								
Low	Low	664	620	588	22	7.68	298	0.35
Low	MidLow	886	853	812	24	7.58	518	0.39
Low	MidHi	1214	1192	1135	28	7.52	849	0.46
Low	High	2066	2077	1975	37	7.39	1957	0.61
MidLow	Low	962	870	813	49	7.90	251	0.80
MidLow	MidLow	1300	1224	1156	52	7.78	474	0.86
MidLow	MidHi	1758	1693	1607	59	7.69	810	0.97
MidLow	High	2984	2976	2834	65	7.48	2292	1.07
MidHi	Low	1444	1278	1166	104	8.07	244	1.70
MidHi	MidLow	1903	1769	1653	100	7.90	512	1.63
MidHi	MidHi	2674	2535	2383	124	7.84	853	2.03
MidHi	High	4339	4261	4053	129	7.63	2353	2.12
High	Low	2181	1848	1604	237	8.29	201	3.88
High	MidLow	2849	2556	2310	232	8.12	427	3.79
High	MidHi	3703	3494	3265	194	7.90	1012	3.19
High	High	6232	6046	5724	236	7.74	2539	3.90
Control		2043	1834	1672	150	8.07	350	2.45
Avg. Brooder		2128	1980	1846	115	7.90	570	1.86
Development		2076	1933	1803	111	7.90	570	1.80

stocked at a density of two embryos mL^{-1} (Experiment 1) or four embryos mL^{-1} (Experiment 2) in both BODs and open culture buckets. Antibiotics were added to all treatments and controls (2 ppm chloramphenicol and 10 ppm ampicillin) before stocking the developing embryos. Prelim-

inary experiments indicated that the antibiotics had no negative effect on the ability of larvae to develop normally, as measured here. *O. lurida* larvae were incubated at 20°C for 5 d before being removed and sampled alongside brooded larvae.

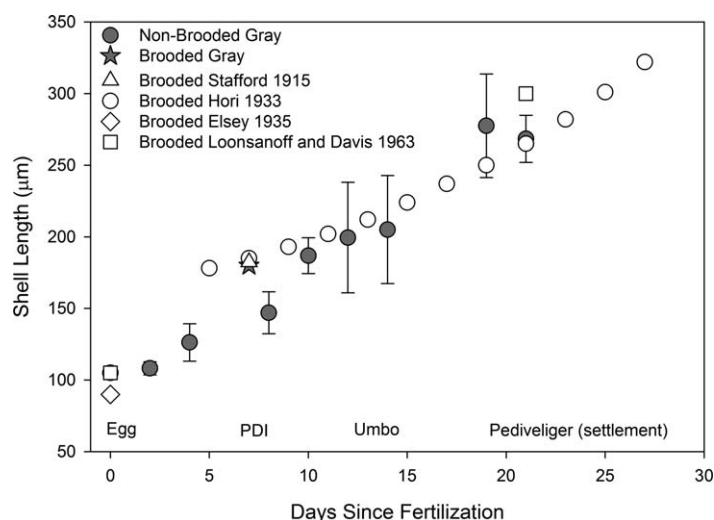


Fig. 2. Sizes of brooded and non-brooded *O. lurida* larvae from preliminary studies and other published studies. Non-brooded larvae were extracted by the techniques outlined in the materials and methods, and reared to settlement. “Brooded Gray” were larvae collected at time of release by *O. lurida*, and other size values were taken from the references noted in the figure caption. Text labels along the x-axis indicate approximate stages during the course of larval growth and development.

Five days after the beginning of the incubation water chemistry samples were taken, and then larvae were rinsed from BOD bottles using water from their respective treatments. Triplicate sub-samples and single samples (~120 larvae per sample) were collected from each BOD bottle, in experiment #2 and #1 respectively, and preserved with 0.1% buffered formalin solution. Due to sampling error, seven treatments (bolded in Table 1) were sampled only once in Experiment 2. Little variance in larval development responses (shell height and normality) was found in the triplicate sub-samples here and in previous experiments (Waldbusser et al. 2015a,b). The consistency in these triplicate sub-samples indicate that our sampling and counting techniques were repeatable and robust. Importantly, for all data analyses, replicate sub-samples from each BOD bottle are averaged to prevent pseudo-replication and artificially inflating error degrees of freedom. In other words we have only one datum point per experimental chamber in our analyses.

Analysis of larval development 168 h post-fertilization was conducted following procedures described in (Waldbusser et al. 2015a,b). Briefly, larval samples were transferred to shell vials and analyzed under a Leica DM IL LED inverted microscope at 10X magnification. The proportion of D-shaped normally developed larvae with a straight hinge and presence of tissue inside the shell vs. total number of larvae was quantified (His et al. 1996).

Shell lengths, defined as the longest axis parallel to the shell hinge, of ≥20 larvae per sample were measured on size-calibrated larval images taken with an inverted microscope at 10X magnification and processed with image software

(Image-Pro v.7). Larvae defined as abnormal by our criteria above were excluded from size determinations to prevent bias, as non-normally developed larvae are usually smaller, and their inclusion results in difficulties disentangling experimental effects of development and growth.

A two-way analysis of variance (ANOVA) was used to determine main effects of the experimental design (P_{CO_2} and Ω_{ar}) on normal development and shell length. Assumptions of normality and homoscedasticity were checked by visual inspection of residuals, and with the Shapiro–Wilk test and Hartley’s F-max test, respectively. The results from each experiment were analyzed separately because of minor differences between experiments in chemistry treatment levels. All statistics were run in SAS v9.3 using the mixed model procedure.

Larval development and calcification of *O. lurida* and *Crassostrea gigas*

In June 2014, in parallel to the previously discussed Experiment 2, we tracked the larval development of one group of *O. lurida* until 168 h post-fertilization and one group of *Crassostrea gigas* until 48 h post-fertilization to document differences in development rates to the Prodissoconch I shell (PDI) stage. We measured calcification rates, lipid content, and took samples for Scanning Electronic Microscopy (SEM) and Cross-polarized light microscopy. *O. lurida* were cultured as described in the section above, except they were kept in open-top 30-L polyethylene open containers (chemical conditions noted as “Development” in Table 1), with a 90% water change 3 d following stocking.

Two days after *O. lurida* displayed signs of reproduction, 12 conditioned *C. gigas* adults were brought to HMSC from the Whiskey Creek Shellfish Hatchery (Netarts, Oregon, U.S.A.) and held overnight in a refrigerator at 5°C. The following day, we spawned the Pacific oysters. Briefly, two males and two females were sacrificed and stripped of gametes. Eggs from both females were combined and 1 mL of combined sperm was added and mixed. 90% of eggs were fertilized after 40 min, and sperm was rinsed. Eggs were counted and stocked at a density of 10 larvae/mL in 5 30 L open culture containers. Antibiotics were added.

O. lurida samples for calcium, SEM and light microscopy were collected every 6 h from 48 h post-fertilization until 168 h post-fertilization. Lipid samples were taken at 0 h post-fertilization and daily from 48 h until 168 h post-fertilization. *C. gigas* larvae were sampled every 3 h from 9 h post-fertilization until 48 h post-fertilization for calcium, SEM and light microscopy. Lipid samples were collected at 0 h, 24 h, and 48 h post-fertilization.

Two duplicate samples of ≥50 individuals per sampling point were targeted for calcium measurements for each species. The exact number of live individuals in each sample was quantified under a Leica S6D dissecting scope before preservation. The larvae were filtered onto Whatman 8.0 μm

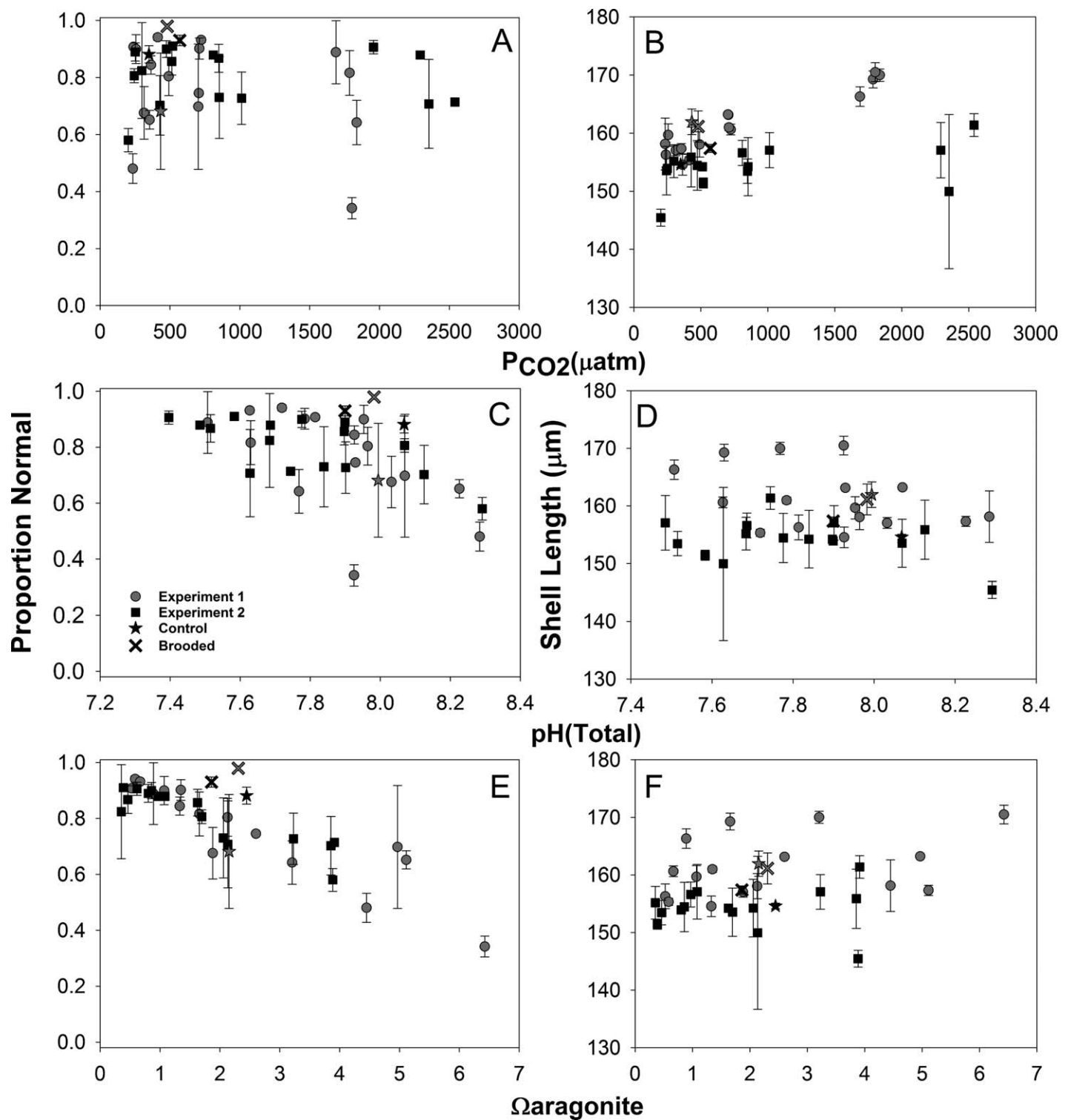


Fig. 3. The response of *O. lurida* shell development and growth of normal shells to P_{CO_2} (A, B), pH (C, D), and Ω_{ar} (E, F), respectively. Control larvae were extracted from brooding females and reared in non-manipulated seawater. "Brooded" larvae were brooded by females and extracted at the end of the experiment and were at the same developmental stage as the previously extracted larvae.

Nucleopore Track-Etch filter membranes and then rinsed with 20 mL of deionized water (DI) to lyse cells. Each dry filter was then placed in a labeled, acid-cleaned 1.5 mL centrifuge tube and immediately frozen at -20°C . Blanks of filtered seawater samples were taken every 6–12 h following the same procedure. Prior to analysis, filters were submerged in 1.8 mL distilled 1.0 M HNO_3 with sonication for 2 min to assure dissolution of the samples. Samples were analyzed on the Tele-dyne Leeman Prodigy Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES) in the Keck Collaboratory at Oregon State University. The average procedural blank ($n = 17$) for calcium was <1 ppb, and we do not report any data under 9 ppb Ca (three times the standard deviation on this average). Accuracy and precision, based on analysis of NIST 1643e standard was $<5\%$, less than the errors reported here that represent the true reproducibility of multiple samples individually processed.

A greater amount of eggs and larvae were needed for lipid measurements. For *O. lurida*, approximately 20,000 eggs or 12,000 larvae were targeted depending of the sample point. For *C. gigas*, 60,000 eggs or 54,000 larvae for *C. gigas* were targeted. The eggs or larvae were filtered onto pre-combusted GF/C grade glass microfiber filters under light vacuum pressure. A coupled blank filter was collected at every sample point and for every species by filtering the same volume of $0.22\ \mu\text{m}$ filtered seawater. All filters were stored at -80°C . Lipids were extracted from homogenized oysters using a chloroform/methanol extraction procedure (Folch et al. 1957; Parrish 1987). Thin layer chromatography with flame ionization detection (TLC/FID) with a MARK V Iatroscan (Iatron Laboratories, Tokyo, Japan) was used to determine lipid classes (wax esters, triacylglycerols (TAG), free fatty acids, sterols (ST), and polar lipid). The larger sizes of *O. lurida* eggs and larvae require standardization of lipids for comparison with lipid values for *C. gigas*; therefore we presented lipid data in terms of the ratio of TAG, (primarily an energy storage lipid), to ST (a structural lipid that tracks total biomass). This index has been used previously to document lipid nutritional status of oyster larvae (Ben Kheder et al. 2010).

Results

Two-day-old *O. lurida* embryos extracted from maternal broodstock developed normally and completed metamorphosis at rates that were comparable to those of brooded larvae under controlled laboratory conditions (Fig. 2). There appears to be no observable short-term physiological benefit from brooding for this species under non-stressful conditions. Embryos extracted from the brood chamber appeared to develop, and ultimately settle, as normally as brooded larvae. This ability to extract and rear larvae outside the brood chamber allowed us to explore acidification exposure effects on developing embryos while removing any direct maternal effects on carbonate chemistry exposure during brooding.

O. lurida PDI larvae showed no acute negative response of normal shell development or shell length of normal larvae to increasing P_{CO_2} or decreasing Ω_{ar} and pH after the 5-d exposure (Fig. 3). Additionally, larvae maintained in brooding females in flow-through seawater over the course of the experiment were no different from those extracted and exposed to treatment conditions (Fig. 3). A slight negative effect of increasing Ω_{ar} , however, appeared to decrease the proportion of normally developed larvae. A two-way analysis of variance indicated some significant effects, but no interaction, of the primary experimental variables (P_{CO_2} and Ω_{ar}) on shell development and length. Increasing Ω_{ar} was found to have a small, but statistically significant, negative effect on normal shell development in experiments 1 and 2, respectively ($F_{3,41} = 42.95$, $p < 0.0001$, $F_{3,41} = 19.26$, $p < 0.0001$). At the highest Ω_{ar} conditions (> 6) there was a nearly 60% reduction in normally developed larvae, however across a more typical range for the region ($\Omega_{\text{ar}} \sim 2.5$), the decline in normally developed larvae was 10–20%, with significant variance among different treatments at this value. P_{CO_2} only had a statistically significant effect in experiment 1 ($F_{3,41} = 5.95$, $p = 0.0018$), but there was no overall positive or negative trend and, only the high P_{CO_2} treatment was significantly different from the two mid-level treatments.

Shell length showed no statistically significant response to carbonate system variables in experiment 2, but there were small, but statistically significant, positive effects of both P_{CO_2} and Ω_{ar} , respectively, in experiment 1 ($F_{3,38} = 107.88$, $p < 0.0001$, $F_{3,38} = 4.34$, $p = 0.0099$). Shell length was smaller in the lowest Ω_{ar} treatment relative to the mid-high and high treatments, by approximately $3\ \mu\text{m}$ (Fig. 3). However, shell length increased in response to P_{CO_2} and larvae from the high P_{CO_2} treatment were approximately $13\ \mu\text{m}$ longer than from the mid-low and low treatments (Fig. 3B). Importantly, total DIC concentration increases with P_{CO_2} in our treatment structure (Table 1), and this positive shell growth response is likely due to DIC, as documented previously in other bivalve species when Ω_{ar} stress is removed (Thomsen et al. 2015; Waldbusser et al. 2015a).

The completion of PDI shell calcification in *O. lurida* appeared to take at least 24 h from the initial detection of calcium carbonate to a fully formed shell (Fig. 4), much longer than the approximately 6 h in *C. gigas* (Waldbusser et al. 2015a) (Fig. 5). Direct measures of calcification rate by Ca^{2+} incorporation were approximately sevenfold slower in *O. lurida* compared with *C. gigas* over the same developmental stage, with a roughly 50-fold slower decline in the ratio triacylglyceride standardized to structural sterol (TAG/ST) lipids (Fig. 6).

Discussion

The lack of negative acute response of Olympia oyster larvae to decreasing Ω_{ar} was initially surprising and counter to responses of other bivalve larvae in similar experiments

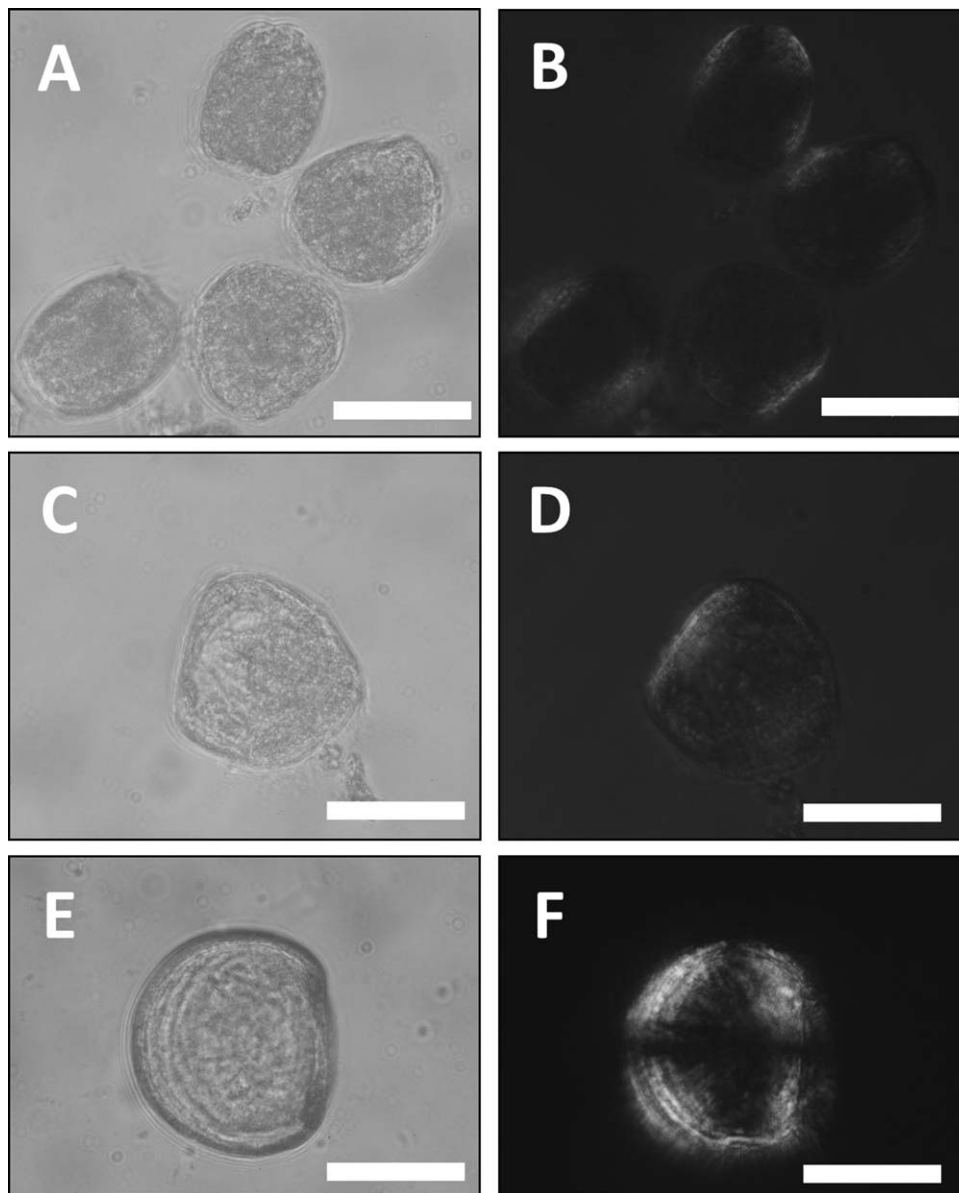


Fig. 4. Early shell development of *O. lurida* at 96 h (A, B, C), 108 h (D, E, F), and 120 h (G, H, I) following fertilization. Time from fertilization is increasing from top (96 h) to bottom (120 h). Images in columns from left to right are, transmission light microscopy (A, D, G), cross-polarized light microscopy (B, E, H), and scanning electron microscopy (C, F, I). All scale bars are 50 μm in length. The glowing regions of the cross-polarized light images indicate calcium carbonate deposition. Due to the curvature of the shell, and secondary refraction of the polarized light, a full calcified shell develops the “Maltese cross,” a dark cross area over the center of the shell that is in fact calcified. The stages are very similar to those documented elsewhere for other larval species (Mouëza et al. 2006; Aranda-Burgos et al. 2014), however the duration from the formation of the periostracum (C) to fully calcified shell (I) takes at least 24 h, as compared to approximately 6 h for *C. gigas* (Fig. 5).

(Gazeau et al. 2011; Thomsen et al. 2015; Waldbusser et al. 2015a,b), all of which are broadcast spawning and rapidly developing. In the absence of evidence for other explanatory physiological mechanisms, we suggest that the lack of acute negative response to low Ω_{ar} coupled to significantly slower PDI calcification rate provides further support for a kinetic-energetic constraint on rapidly calcifying PDI bivalve larvae (Waldbusser et al. a,b). Furthermore, slower development

(and thus slower calcification) may be a mechanism by which brooding indirectly provides increased resiliency to ocean acidification in marine invertebrates as seen in recent studies (Noisette et al. 2014; Lucey et al. 2015). The slower calcification rate during *O. lurida* PDI development coupled to a lack of acute response does not provide an absolute proof of a trait for resiliency to OA; however our results suggest the possibility of slow shell building as an example of

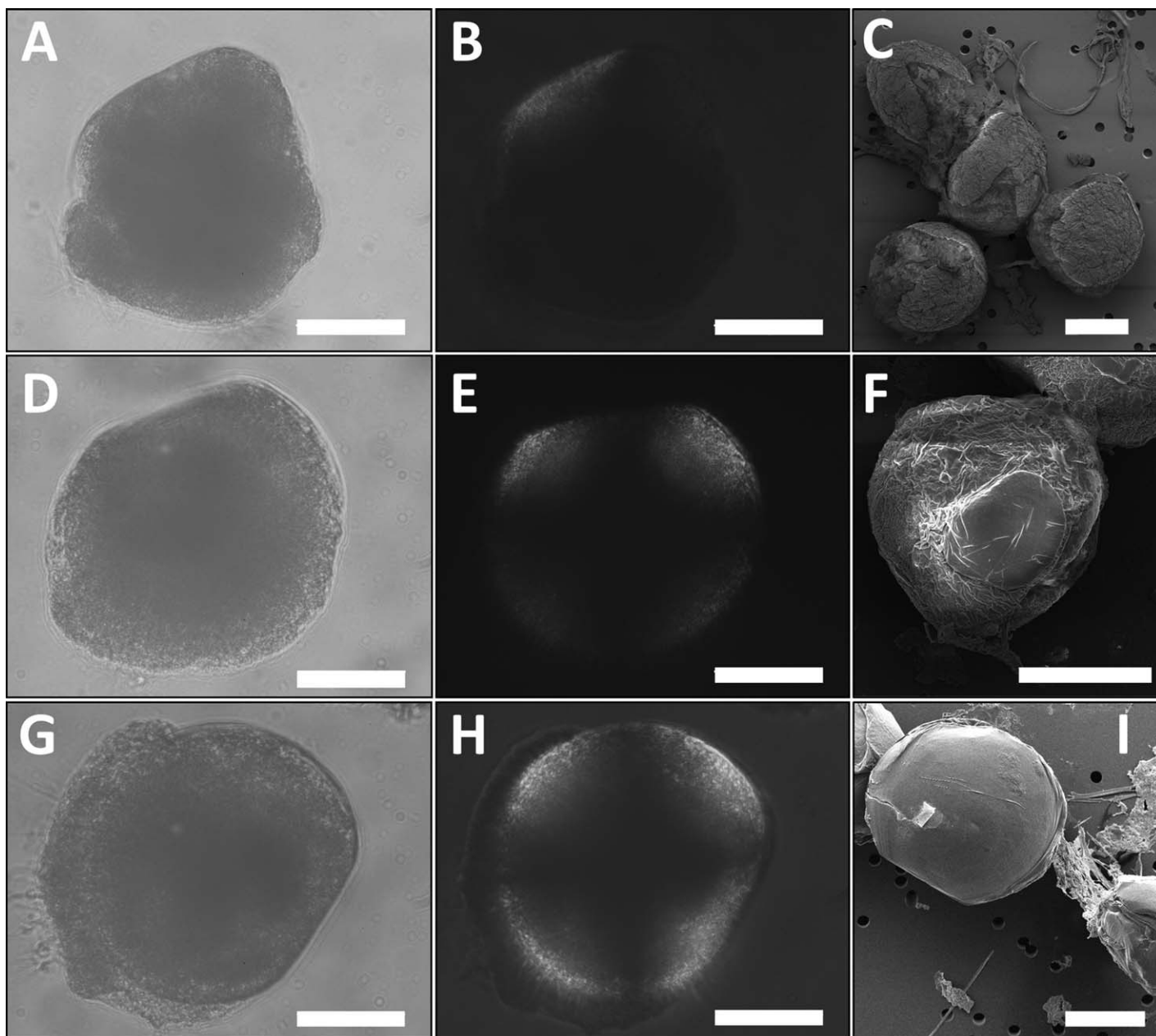


Fig. 5. Transmission (A, C, E) and cross-polarized (B, D, F) light microscope images of Pacific oyster larvae (*C. gigas*) at 18 (A, B), 21 (C, D), and 24 (E, F) h post-fertilization. Time is increasing as in the previous graph from top (18 h) to bottom (24 h). Scale bars are 50 μm .

exaptation (Gould and Vrba 1982). Brooding, as a reproductive strategy, is often cited as a mechanism for reducing the period of the larval stage and consequently decreased predation and increased retention in estuaries (Gaines and Bertness 1992; Sponaugle et al. 2002). Brooding has evolved and is evident in some marine invertebrates for a litany of reasons (Thorson 1950), but most likely not as a direct adaptation to elevated carbon dioxide. Thus if the slower calcification associated with brooding proves to be the mechanism for resiliency to OA, this would highlight how existing traits may

provide some scope for adapting to climate change without requiring genetic selection, a generally slow process.

Several notable observations should be addressed before further discussing the implications of our findings. The linear increase in calcium accumulation for *C. gigas* and *O. lurida* suggests constant calcification rates during this shell building stage, with the exception of two values for the *O. lurida* that were well above the linear trend of the other Ca^{2+} measurements. These calcium values were elevated in duplicate samples, and while highly variable at those two

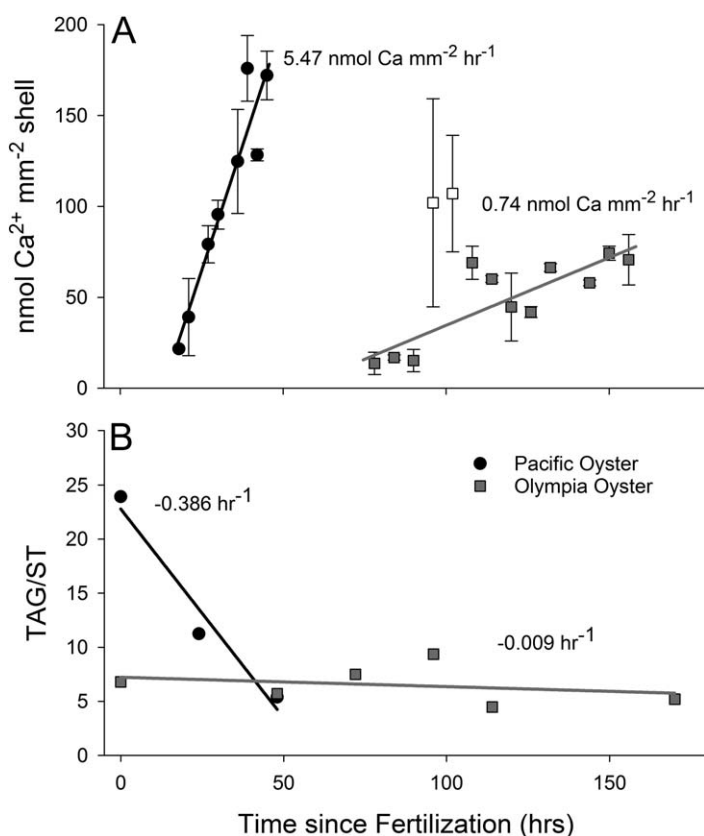


Fig. 6. Calcium concentration in larval samples collected during the period of PDI shell formation in the Pacific and Olympia oysters (A). The concentration has been corrected for the number of larvae in each sample, and the approximate total planar area of the periostracum derived from scanning electron microscope imagery. Rates of calcification were computed from the slope of the linear regression line. Two values for *O. lurida* were excluded for the regression fit (open squares), see text. The ratio of triacylglycerol (TAG) to sterol (ST) lipids over the period of PDI shell formation in both species (B) provides a standardized estimate in primary energy lipids to structural lipids (Ben Kheder et al. 2010).

time points, they appear generally consistent and we thus consider them real. It is possible these two outliers are related to a calcium concentrating mechanism for PDI shell formation, such as the use of CaCO_3 precursors in hemocytes repairing shells of adult oysters (Mount et al. 2004). The several studies of early calcification and shell development in bivalves (LaBarber 1974; Eyster 1986; Medakovic 1997) do not address the exact mechanisms of PDI biocalcification operating at cellular and sub-cellular levels. Nor did we have the temporal resolution or analytical tools to confirm this or identify other possibilities. It is however odd that the overall concentration drops again following these points, so we simply note the anomalous nature of the data points in the overall dataset.

A second interesting finding was that elevated Ω_{ar} resulted in reduced normal shell development (but not shell growth) in developing *O. lurida* larvae; while intriguing, it is notable that values of Ω_{ar} rarely exceed 4 in temperate marine

waters, so the environmental relevance is likely minimal (but still physiologically interesting). Similar studies on other broadcast spawning species have found generally positive effects of Ω_{ar} up to values of ~ 7 (Waldbusser et al. 2015a,b). It is plausible that the elevated Ω_{ar} creates challenges for *O. lurida* in controlling the slower calcification rate while coordinating shell formation with organogenesis.

Finally, while we found no negative acute effect of ocean acidification on developing embryos of *O. lurida*, negative chronic and carry-over effects have been well documented in post-release, later stage larvae and juveniles (Hettinger et al. 2012, 2013). Recognition of different OA responses across life-history stages within a species suggests that we must begin to understand better the alignment of critical life history stages with periods of increased environmental variability or reduced windows of favorable conditions associated with global climate change (Waldbusser and Salisbury 2014; Hales et al. unpubl.). This nuanced view often is difficult to capture in large meta-analyses, or with broad statements of whether a given species is sensitive to OA. For the current study the critical point is the acute and ubiquitous OA effects on early development documented for other broadcast spawning oyster and mussel embryos (Gazeau et al. 2011; Thomsen et al. 2015; Waldbusser et al. 2015a,b) were not observed in *O. lurida* at this critical life-history stage.

The comparison between the native *O. lurida* and the introduced *C. gigas* begs a brief discussion of the chemical habitats from which they originally evolved. The *C. gigas* broodstock used throughout the U.S. west coast industry are derived primarily from the Miyagi region of Japan waters and were initially imported through British Columbia (Quayle 1988). Current day carbonate chemistry conditions in surface waters of the Kuroshio current and associated coastal waters tend to be less dynamic and closer to atmospheric P_{CO_2} levels (Ishii et al. 2011; Yamamoto-Kawai et al. 2015) than the upwelling dominated conditions located in the East Pacific/California Current and Oregon's coastal waters (Feely et al. 2008; Barton et al. 2012; Harris et al. 2013). One would therefore assume *O. lurida* to have greater physiological tolerance and adaptations to cope with the highly dynamic and often poor carbonate chemistry conditions located in Oregon's waters, and this doesn't explicitly have to be related to slower development and calcification. From a strictly physiological-environmental tolerance perspective (e.g., Pörtner 2008), other pairs of bivalve species (located in our upwelling system vs. other more stable systems) should, in theory, follow suit. In contrast a comparison of native and introduced mussels to Oregon's coast suggests otherwise, the response of *Mytilus californianus* larvae to Ω_{ar} was nearly identical to that of the Mediterranean mussel, *Mytilus galloprovincialis* (Waldbusser et al. 2015b). Additionally, the native *M. californianus* grew larger and developed normally at elevated Ω_{ar} experimental conditions (Waldbusser et al. 2015b), counter to the findings here for *O.*

lurida. We recognize our findings are suggestive but not conclusive, and we also highlight that a kinetic-energy based sensitivity related to calcification rates of the PDI shell (Waldbusser et al. 2013) does not have to be mutually exclusive of physiological tolerance related to adaptation in specific environmental conditions (Portner 2008; Waldbusser et al. 2015c).

One important consequence of brooding in *Ostrea* spp. is a typically slower development through the PDI phase (Hopkins 1936; Baker 1995; Medakovic et al. 1997), with the transition from PDI to PDII occurring between 5 d and 7 d, vs. at ~ 2 d in *Crassostrea* spp. The slower development and calcification is visualized in Figs. 4, 5, indicating both a later onset of calcification in *O. lurida*, and a longer period to complete a fully enclosed shell. In our kinetic-energetic hypothesis for oyster larval sensitivity to saturation state (Waldbusser et al. 2013), we argue the constraint can be described by the general kinetic equation for calcium carbonate precipitation:

$$r = k(\Omega - 1) \quad (1)$$

where r is the calcification rate, k is the rate constant, and Ω is the saturation state for the calcium carbonate mineral (we assume the reaction order is 1 for simplicity), although it is important to point out that this physical-chemical representation of the rate can be modified in important ways by the mediating organisms. Specifically, we previously argued that at a fixed (and rapid) calcification rate (r), k must increase dramatically as ambient-water Ω_{ar} approaches undersaturation (or unity), and thus energy must be put into the system to elevate k at the precipitation site (Waldbusser et al. 2013, 2015c). Alternatively, organisms may employ the use of cross-membrane ion pumps to elevate Ω at the precipitation site. While the exact mechanisms are still unknown, either scenario requires increasing energetic demand as ambient-water Ω_{ar} drops. It appears in the case of *O. lurida*, and possibly other brooded, slowly-calcifying bivalves, that the energetic stress induced by ocean acidification is avoided by a lower rate (r), and thus the energetic demand is reduced. In other words, a slower calcification rate, as seen in Fig. 6, results in a subsequent lower relative energetic burden through initial shell formation (Fig. 6B). This finding is consistent with our finding of a lower relative maternal investment in energy lipids for this species, compared to the rapidly developing *C. gigas* (note the roughly 5x greater TAG/ST in *O. lurida*). So while the eggs are larger in *O. lurida*, the investment is not in energy lipids, but more likely structural components such as sterols and protein. Biocalcification rates measured here are, however, still far above abiotic CaCO_3 precipitation rates (Maeda-Martinez 1987; Zuddas and Mucci 1994), and thus should require energetic investment to accelerate. Better quantification of the energetic demand of accelerating biocalcification is sorely needed to understand this mechanism, however recent work on sea

urchin larvae suggests the greater change in energy allocation under OA stress is associated with protein synthesis and turnover (Pan et al. 2015). Changes in the protein composition or content of larval shell could also be a compensatory response in bivalve larvae; however only changes in the synthesis or turnover of protein was seen in sea urchin larvae (Pan et al. 2015) but no differences were found in total protein content.

Inquiry into the ecological trade-offs between brooding and broadcast spawning by marine invertebrates spans many decades (Thorson 1950; Strathmann 1985; Pechenik 1999). Our experiments indicate that, under non-stressful conditions, brooding in the Olympia oyster conveys no direct physiological or developmental benefit for the developing embryos and larvae (Fig. 2). As others have previously debated, the brooding behavior likely conveys fitness at ecological scales by reducing pelagic larval duration, thus reducing pelagic predation and increasing local retention of larvae (Wray 1995; Sponaugle et al. 2002). Brooding behavior typically results in larger maternal investment per egg, with fewer larvae than broadcast spawners, and this holds true in the comparison between *O. lurida* and *C. gigas*. Olympia oyster eggs are typically twice the diameter, and larvae at settlement are typically greater than 30% larger than *C. gigas*; however, adult sizes are the inverse, with *C. gigas* adults growing several times larger than *O. lurida*. While it is generally agreed that brooding results in greater maternal investment per egg (Jaeckle 1995), and total lipids were absolutely higher in the larger *O. lurida* eggs, the relative investment into energy lipids (TAG/ST) is much lower in *O. lurida* than *C. gigas* (Fig. 6). These results suggest that the slower development (and thus also calcification) in *O. lurida* is relatively less energetically expensive, and supports our hypothesis of the rapid calcification during PDI in broadcast spawners (or slower calcification in this brooder) playing an important role in the sensitivity of this life-history stage of bivalves.

Recent experiments and analyses indicate that brooding in calcifiers and non-calcifying species appears to convey some fitness to ocean acidification (Noisette et al. 2014; Lucey et al. 2015); therefore slower calcification may only be one trait of several that provides brooded oyster larvae resistance to acute OA effects during PDI development. Conversely, brooding may also create adverse conditions for larvae; for example, extended shell closure due to stress in the oyster *Ostrea chilensis* during brooding can generate low pH after several hours (Chaparro et al. 2009; Segura et al. 2015). During normal ventilation, however, seawater is rapidly flushed through the brood chamber suggesting there is little time for respiratory carbon dioxide accumulation (Chaparro et al. 1993; Mardones-Toledo et al. 2015) and salinity events that cause *O. chilensis* to “clam up,” in Chile are uncommon during the brooding period (Segura et al. 2015). Our results suggest a fairly high threshold for larval tolerance to less favorable carbonate chemistry (low pH/ Ω), but longer term, more chronic effects on post-release *O. lurida* larvae

have been noted elsewhere (Hettinger et al. 2012). Another possibility for resistance to OA is acclimatization from bicarbonate accumulation, an often noted as a key mechanism to cope with acidification on shorter time scales (Pane and Barry 2007; Melzner et al. 2009). This acclimatization however generally requires several days (Pane and Barry 2007; Melzner et al. 2009), and is unlikely here as larvae were not brooded. More work is needed to understand the interplay of brooding, environmental triggers of extended valve closure, and whether larval fitness is provided through acclimation to elevated CO₂ in a semi-isolated brood chamber (our results suggest not).

Natural selection, the agent of evolution, selects for traits that provide fitness to species in their habitat and, therefore, ultimately a trait is needed to explain the OA resiliency we documented in Olympia oyster embryos. We propose that this trait may be slow shell building, which counteracts the well-documented OA related constraint of decreased saturation state (Gazeau et al. 2011; Waldbusser et al. 2013; Thomsen et al. 2015). Therefore, exaptation (Gould and Vrba 1982) could be cited as one explanation for the lack of acute response seen in Olympia oyster embryos and larvae, as the slow-shell building trait is a side-effect of brooding. The possibility of coopting current traits for fitness has rarely been recognized in the ocean acidification literature, and our findings suggest that some species may have traits conveying fitness to counter OA stress already at their disposal. We however acknowledge that more specific studies across a wider range of species would be needed to provide the inferential power needed, but we highlight this here to spur interest in exploring this neglected component of evolutionary ecology in understanding potential for rapid adaptation in marine organisms to OA. Given the significance of the PDI bottleneck for success in bivalve larvae, and role it has played in the oyster seed shortage in the U.S. Pacific Northwest (Barton et al. 2012, 2015), identifying this acute fitness in the native Olympia oyster provides some hope in the near-term for persistence of this species and perhaps brooding species in general (Lucey et al. 2015). The potential for exaptation also optimistically means that some organisms may already have traits that could provide fitness in a high CO₂ world and thus may not require the long multi-generational time-scales needed for selection and development of new traits resulting directly from ocean acidification exposure.

References

- Aranda-Burgos, J. A., F. Da Costa, S. Nóvoa, J. Ojea, and D. Martínez-Patiño. 2014. Embryonic and larval development of *Ruditapes decussatus* (Bivalvia: Veneridae): A study of the shell differentiation process. *J. Molluscan Stud.* **80**: 8–16. doi:10.1093/mollus/eyt044
- Baker, P. 1995. Review of ecology and fishery of the Olympia oyster, *Ostrea lurida* with annotated bibliography. *J. Shellfish Res.* **14**: 501–518.
- Bandstra, L., B. Hales, and T. Takahashi. 2006. High-frequency measurements of total CO₂: Method development and first oceanographic observations. *Mar. Chem.* **100**: 24–38. doi:10.1016/j.marchem.2005.10.009
- Barton, A., B. Hales, G. G. Waldbusser, C. Langdon, and R. A. Feely. 2012. The Pacific oyster, *Crassostrea gigas*, shows negative correlation to naturally elevated carbon dioxide levels: Implications for near-term ocean acidification effects. *Limnol. Oceanogr.* **57**: 698–710. doi:10.4319/lo.2012.57.3.0698
- Barton, A., and others. 2015. Impacts of coastal acidification on the Pacific Northwest shellfish industry and adaptation strategies implemented in response. *Oceanography* **28**: 146–159. doi:10.5670/oceanog2015.38
- Ben Kheder, R., C. Quere, J. Moal, and R. Robert. 2010. Effect of nutrition on *Crassostrea gigas* larval development and the evolution of physiological indices: Part B: Effects of temporary food deprivation. *Aquaculture* **308**: 174–182. doi:10.1016/j.aquaculture.2010.08.030
- Byrne, M. 2011. Impact of ocean warming and ocean acidification on marine invertebrate life history stages: Vulnerabilities and potential for persistence in a changing ocean. *Oceanogr. Mar. Biol. Annu. Rev.* **49**: 1–42. doi:10.1201/b11009-2
- Chaparro, O. R., R. J. Thompson, and J. E. Ward. 1993. In vivo observations of larval brooding in the Chilean oyster, *Ostrea chilensis* Philippi, 1845. *Biol. Bull.* **185**: 365–372. doi:10.2307/1542477
- Chaparro, O. R., C. J. Segura, J. A. Montory, J. M. Navarro, and J. A. Pechenik. 2009. Brood chamber isolation during salinity stress in two estuarine mollusk species: From a protective nursery to a dangerous prison. *Mar. Ecol. Prog. Ser.* **374**: 145–155. doi:10.3354/meps07777
- Dickson, A. G. 1990. Standard potential of the reaction: AgCl (s) + 12H₂ (g) = Ag (s) + HCl (aq), and the standard acidity constant of the ion HSO₄[−] in synthetic sea water from 273.15 to 318.15 K. *J. Chem. Thermodyn.* **22**: 113–127. doi:10.1016/0021-9614(90)90074-Z
- Doney, S. C., V. J. Fabry, R. A. Feely, and J. A. Kleypas. 2009. Ocean acidification: The other CO₂ problem. *Mar. Sci.* **1**: 169–192. doi:10.1146/annurev.marine.010908.163834
- Eyster, L. S. 1986. Shell inorganic composition and onset of shell mineralization during bivalve and gastropod embryogenesis. *Biol. Bull.* **170**: 211–231. doi:10.2307/1541804
- Feely, R. A., C. L. Sabine, J. M. Hernandez-Ayon, D. Ianson, and B. Hales. 2008. Evidence for upwelling of corrosive “acidified” water onto the continental shelf. *Science* **320**: 1490–1492. doi:10.1126/science.1155676
- Feely, R. A., and others. 2010. The combined effects of ocean acidification, mixing, and respiration on pH and carbonate saturation in an urbanized estuary. *Estuar. Coast. Shelf Sci.* **88**: 442–449. doi:10.1016/j.ecss.2010.05.004
- Foden, W. B., and others. 2013. Identifying the world’s most climate change vulnerable species: A systematic trait-

- based assessment of all birds, amphibians and corals. *PLoS ONE* **8**: e65427. doi:[10.1371/journal.pone.0065427](https://doi.org/10.1371/journal.pone.0065427)
- Folch, J., and others. 1957. A simple method for the isolation and purification of total lipids from animal tissues. *J. Biol. Chem.* **226**: 497–509.
- Gaines, S. D., and M. D. Bertness. 1992. Dispersal of juveniles and variable recruitment in sessile marine species. *Nature* **360**: 579–580. doi:[10.1038/360579a0](https://doi.org/10.1038/360579a0)
- Gazeau, F., J.-P. Gattuso, M. Greaves, H. Elderfield, J. Peene, C. H. Heip, and J. J. Middelburg. 2011. Effect of carbonate chemistry alteration on the early embryonic development of the Pacific oyster (*Crassostrea gigas*). *PloS One* **6**: e23010. doi:[10.1371/journal.pone.0023010](https://doi.org/10.1371/journal.pone.0023010)
- Gazeau, F., L. M. Parker, S. Comeau, J. P. Gattuso, W. A. O'Conner, S. Martin, H-O. Pörtner, and P.M. Ross. 2013. Impact of ocean acidification on marine shelled molluscs. *Marine Biology* **160**: 2207–2245. doi:[10.1007/s00227-013-2219-3](https://doi.org/10.1007/s00227-013-2219-3)
- Gould, S. J., and E. S. Vrba. 1982. Exaptation—a missing term in the science of form. *Paleobiology* **8**: 4–15. doi:[10.1017/S0094837300004310](https://doi.org/10.1017/S0094837300004310)
- Hales, B., T. Takahashi, and L. Bandstra. 2005. Atmospheric CO₂ uptake by a coastal upwelling system. *Global Biogeochem. Cycles* **19**. doi:[10.1029/2004GB002295](https://doi.org/10.1029/2004GB002295)
- Harris, K. E., M. D. DeGrandpre, and B. Hales. 2013. Aragonite saturation state dynamics in a coastal upwelling zone. *Geophys. Res. Lett.* **40**: 2720–2725. doi:[10.1002/grl.50460](https://doi.org/10.1002/grl.50460)
- Hauri, C., N. Gruber, A. M. P. McDonnell, and M. Vogt. 2013. The intensity, duration, and severity of low aragonite saturation state events on the California continental shelf. *Geophys. Res. Lett.* **40**: 3424–3428. doi:[10.1002/grl.50618](https://doi.org/10.1002/grl.50618)
- Hettinger, A., and others. 2012. Persistent carry-over effects of planktonic exposure to ocean acidification in the Olympia oyster. *Ecology* **93**: 2758–2768. doi:[10.1890/12-0567.1](https://doi.org/10.1890/12-0567.1)
- Hettinger, A., E. Sanford, T. M. Hill, J. D. Hofelt, A. D. Russell, and B. Gaylord. 2013. The influence of food supply on the response of Olympia oyster larvae to ocean acidification. *Biogeosci. Discuss.* **10**: 5781–5802. doi:[10.5194/bgd-10-5781-2013](https://doi.org/10.5194/bgd-10-5781-2013)
- His, E., M. N. L. Seaman, and R. Beiras. 1996. A simplification the bivalve embryogenesis and larval development bioassay method for water quality assessment. *Water Res.* **31**: 351–355. doi:[10.1016/s0043-1354\(96\)00244-8](https://doi.org/10.1016/s0043-1354(96)00244-8)
- Hopkins, A. E. 1936. Ecological observations on spawning and early larval development in the Olympia oyster (*Ostrea Lurida*). *Ecology* **17**: 551–566. doi:[10.2307/1932760](https://doi.org/10.2307/1932760)
- IPCC. 2014. Climate Change 2014: Impacts, adaptation, and vulnerability, 1132 p. Part A: Global and sectoral aspects. In C. B. Field and others [eds.], Contribution of Working Group II to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge Univ. Press.
- Ishii, M., N. Kosugi, D. Sansano, S. Saito, T. Midorikawa, and H. Y. Inoue. 2011. Ocean acidification off the south coast of Japan: A result from time series observations of CO₂ parameters from 1994 to 2008. *J. Geophys. Res.* **116**: C06022. doi:[10.1029/2010JC006831](https://doi.org/10.1029/2010JC006831)
- Jaekle, W. 1995. Variation in the size, energy content, and biochemical composition of invertebrate eggs: Correlates to the mode of larval development, p. 49–78. In L. McEdward [ed.] *Ecology of marine invertebrate larvae*. CRC Press, Boca Raton, FL.
- LaBarber, M. 1974. Calcification of first larval shell of *Tridacna squamosa* (Tridacnidae-Bivalvia). *Mar. Biol.* **25**: 233–238. doi:[10.1007/bf00394969](https://doi.org/10.1007/bf00394969)
- Lucey, N. M., C. Lombardi, L. DeMarchi, A. Schulze, M. C. Gambi, and P. Calosi. 2015. To brood or not to brood: Are marine invertebrates that protect their offspring more resilient to ocean acidification? *Sci. Rep.* **5**. doi:[10.1038/srep12009](https://doi.org/10.1038/srep12009)
- Lueker, T. J., A. Dickson, and C. D. Keeling. 2000. Ocean pCO₂ calculated from dissolved inorganic carbon, alkalinity, and equations for K₁ and K₂: validation based on laboratory measurements of CO₂ in gas and seawater at equilibrium. *Marine Chemistry* **70**: 105–119. doi:[10.1016/S0304-4203\(00\)00022-0](https://doi.org/10.1016/S0304-4203(00)00022-0)
- Lüthi, D., and others. 2008. High-resolution carbon dioxide concentration record 650,000–800,000 years before present. *Nature* **453**: 379–382. doi:[10.1038/nature06949](https://doi.org/10.1038/nature06949)
- Maeda-Martinez, A. 1987. The rates of calcium deposition in shells of molluscan larvae. *Comp. Biochem. Physiol. A Physiol.* **86**: 21–28. doi:[10.1016/0300-9629\(87\)90270-2](https://doi.org/10.1016/0300-9629(87)90270-2)
- Mardones-Toledo, D. A., and others. 2015. Brooding in the Chilean oyster *Ostrea chilensis*: Unexpected complexity in the movements of brooded offspring within the mantle cavity. *PloS One* **10**. doi:[10.1371/journal.pone.0122859](https://doi.org/10.1371/journal.pone.0122859)
- Medakovic, D., S. Popovic, B. Grzeta, M. Plazonic, and M. Hrs-Brenko. 1997. X-ray diffraction study of calcification processes in embryos and larvae of the brooding oyster *Ostrea edulis*. *Mar. Biol.* **129**: 615–623. doi:[10.1007/s002270050204](https://doi.org/10.1007/s002270050204)
- Melzner, F., and others. 2009. Physiological basis for high CO₂ tolerance in marine ectothermic animals: Pre-adaptation through lifestyle and ontogeny? *Biogeosciences* **6**: 2313–2331. doi:[10.5194/bg-6-2313-2009](https://doi.org/10.5194/bg-6-2313-2009)
- Millero, F. J. 1995. Thermodynamics of the carbon dioxide system in the oceans. *Geochim. Cosmochim. Acta* **59**: 661–677. doi:[10.1016/0016-7037\(94\)00354-O](https://doi.org/10.1016/0016-7037(94)00354-O)
- Millero, F.J. 2010. Carbonate constants for estuarine waters, *Mar. Freshwater Res.* **61**: 139–142. doi:[10.1071/MF09254](https://doi.org/10.1071/MF09254)
- Mouëza, M., O. Gros, and L. Frenkiel. 2006. Embryonic development and shell differentiation in *Chione cancellata* (Bivalvia, Veneridae): An ultrastructural analysis. *Invertebr. Biol.* **125**: 21–33. doi:[10.1111/j.1744-7410.2006.00036.x](https://doi.org/10.1111/j.1744-7410.2006.00036.x)
- Mount, A. S., A. P. Wheeler, R. P. Paradkar, and D. Snider. 2004. Hemocyte-mediated shell mineralization in the

- eastern oyster. *Science* **304**: 297–300. doi:[10.1126/science.1090506](https://doi.org/10.1126/science.1090506)
- Noiset, F., and others. 2014. Does encapsulation protect embryos from the effects of ocean acidification? The example of *Crepidula fornicata*. *PLoS ONE* **9**: e93021. doi:[10.1371/journal.pone.0093021](https://doi.org/10.1371/journal.pone.0093021)
- Pan, T. C. F., S. L. Applebaum, and D. T. Manahan. 2015. Experimental ocean acidification alters the allocation of metabolic energy. *Proc. Natl. Acad. Sci.* **112**: 4696–4701. doi:[10.1073/pnas.1416967112](https://doi.org/10.1073/pnas.1416967112)
- Pane, E. F., and J. P. Barry. 2007. Extracellular acid-base regulation during short-term hypercapnia is effective in a shallow-water crab, but ineffective in a deep-sea crab. *Mar. Ecol. Prog. Ser.* **334**: 1–9. doi:[10.3354/meps334001](https://doi.org/10.3354/meps334001)
- Parrish, C. C. 1987. Separation of aquatic lipid classes by chromarod thin-layer chromatography with measurement by latroscan flame ionization detection. *Can. J. Fish. Aquat. Sci.* **44**: 722–731. doi:[10.1139/f87-087](https://doi.org/10.1139/f87-087)
- Pechenik, J. A. 1999. On the advantages and disadvantages of larval stages in benthic marine invertebrate life cycles. *Mar. Ecol. Prog. Ser.* **177**: 269–297. doi:[10.3354/meps177269](https://doi.org/10.3354/meps177269)
- Phillips, B. L., G. P. Brown, and R. Shine. 2010. Life-history evolution in range-shifting populations. *Ecology* **91**: 1617–1627. doi:[10.1890/09-0910.1](https://doi.org/10.1890/09-0910.1)
- Pörtner, H.-O. 2008. Ecosystem effects of ocean acidification in times of ocean warming: A physiologist's view. *Mar. Ecol. Prog. Ser.* **373**: 203–217. doi:[10.3354/meps07768](https://doi.org/10.3354/meps07768)
- Quayle, D. B. 1988. Pacific oyster culture in British Columbia. *Can. B. Fish. Aquat. Sci.* **218**: 241.
- Sammarco, P. W., and K. B. Strychar. 2009. Effects of climate change/global warming on coral reefs: Adaptation/exaptation in corals, evolution in zooxanthellae, and biogeographic shifts. *Environ. Bioindic.* **4**: 9–45. doi:[10.1080/15555270902905377](https://doi.org/10.1080/15555270902905377)
- Segura, C. J., J. A. Montory, V. M. Cubillos, C. M. Diederich, J. A. Pechenik, and O. R. Chaparro. 2015. Brooding strategy in fluctuating salinity environments: Oxygen availability in the pallial cavity and metabolic stress in females and offspring in the Chilean oyster *Ostrea chilensis*. *J. Comp. Physiol. B* **185**: 659–668. doi:[10.1007/s00360-015-0908-6](https://doi.org/10.1007/s00360-015-0908-6)
- Sponaugle, S., and others. 2002. Predicting self-recruitment in marine populations: Biophysical correlates and mechanisms. *Bull. Mar. Sci.* **70**: 341–375.
- Strathmann, R. R. 1985. Feeding and nonfeeding larval development and life-history evolution in marine invertebrates. *Annu. Rev. Ecol. Syst.* **16**: 339–361. doi:[10.1146/annurev.es.16.110185.002011](https://doi.org/10.1146/annurev.es.16.110185.002011)
- Takahashi, T., and others. 2014. Climatological distributions of pH, pCO₂, total CO₂, alkalinity and CaCO₃ saturation in the global surface ocean, and temporal changes at selected locations. *Mar. Chem.* **164**: 95–125. doi:[10.1016/j.marchem.2014.06.004](https://doi.org/10.1016/j.marchem.2014.06.004)
- Thomsen, J., K. Haynert, K. M. Wegner, and F. Melzner. 2015. Impact of seawater carbonate chemistry on the calcification of marine bivalves. *Biogeosci. Discuss.* **12**: 1543–1571. doi:[10.5194/bgd-12-1543-2015](https://doi.org/10.5194/bgd-12-1543-2015)
- Thorson, G. 1950. Reproductive and larval ecology of marine bottom invertebrates. *Biol. Rev.* **25**: 1–45. doi:[10.1111/j.1469-185X.1950.tb00585.x](https://doi.org/10.1111/j.1469-185X.1950.tb00585.x)
- Waldbusser, G. G., H. Bergschneider, and M. A. Green. 2010. Size-dependent pH effect on calcification in post-larval hard clam *Mercenaria* spp. *Marine Ecological Progress Series* **417**: 177–182.
- Waldbusser, G. G., E. P. Voigt, H. Bergschneider, M. A. Green, and R. I. E. Newell. 2011. Biocalcification in the Eastern Oyster (*Crassostrea virginica*) in relation to long-term trends in Chesapeake Bay pH. *Estuaries Coast.* **34**: 221–231. doi:[10.1007/s12237-010-9307-0](https://doi.org/10.1007/s12237-010-9307-0)
- Waldbusser, G. G., E. L. Brunner, B. A. Haley, B. Hales, C. J. Langdon, and F. G. Prahl. 2013. A developmental and energetic basis linking larval oyster shell formation to acidification sensitivity. *Geophys. Res. Lett.* **40**: 2171–2176. doi:[10.1002/grl.50449](https://doi.org/10.1002/grl.50449)
- Waldbusser, G. G., and J. E. Salisbury. 2014. Ocean Acidification in the Coastal Zone from an Organism's Perspective: Multiple System Parameters, Frequency Domains, and Habitats, p. 221–247. *Annual Review of Marine Science*, Vol 6. Annual Review of Marine Science.
- Waldbusser, G. G., and others. 2015a. Saturation-state sensitivity of marine bivalve larvae to ocean acidification. *Nat. Clim. Change* **5**: 273–280. doi:[10.1038/nclimate2479](https://doi.org/10.1038/nclimate2479)
- Waldbusser, G. G., and others. 2015b. Ocean acidification has multiple modes of action on bivalve larvae. *PLoS ONE* **9**: e0128376. doi:[10.1371/journal.pone.0128376](https://doi.org/10.1371/journal.pone.0128376)
- Waldbusser, G. G., B. Hales, and B. A. Haley. 2015c. Comment: Calcium carbonate saturation state: on myths and this or that stories. *ICES J. Mar. Sci.* **73**: 563–568. doi:[10.1093/icesjms/fsv174](https://doi.org/10.1093/icesjms/fsv174)
- Wethey, D. S., S. A. Woodin, T. J. Hilbish, S. J. Jones, F. P. Lima, and P. M. Brannock. 2011. Response of intertidal populations to climate: Effects of extreme events versus long term change. *J. Exp. Mar. Biol. Ecol.* **400**: 132–144. doi:[10.1016/j.jembe.2011.02.008](https://doi.org/10.1016/j.jembe.2011.02.008)
- Wray, G. A. 1995. Evolution of larvae and developmental modes. *Ecol. Mar. Invertebr. Larvae* In L. McEdward[ed.] *Ecology of marine invertebrate larvae*. CRC Press, Boca Raton, FL, p 413–447.
- Yamamoto-Kawai, M., N. Kawamura, T. Ono, A. Kubo, M. Ishii, and J. Kanda. 2015. Calcium carbonate saturation and ocean acidification in Tokyo Bay, Japan. *J. Oceanogr.* **71**: 427–439. doi:[10.1007/s10872-015-0302-8](https://doi.org/10.1007/s10872-015-0302-8)
- Zuddas, P., and A. Mucci. 1994. Kinetics of calcite precipitation from seawater: I. A classical chemical kinetics description for strong electrolyte solutions. *Geochim. Cosmochim. Acta* **58**: 4353–4362. doi:[10.1016/0016-7037\(94\)90339-5](https://doi.org/10.1016/0016-7037(94)90339-5)

Acknowledgments

The authors would like to thank Oregon Oyster Company for providing *Ostrea lurida* broodstock, N. Danilchik for imaging bivalve larvae, K. Kovalchik for working on larval calcium measurements, and L. Copeman for running lipid samples and advice on interpretation. A previous version of this manuscript was improved by comments from A. Hettinger, S.E. Kolesar, one anonymous reviewer, and the Associate Editor, James Leichter. This work was supported by National Science Foundation OCE CRI-OA #1041267 to GGW, BH, CJL, and BAH.

Submitted 5 September 2015

Revised 16 March 2016

Accepted 2 May 2016

Associate editor: James Leichter